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ABSTRACT

Among various topology optimization methods used in fluid flow problems, density approach has gained more interest compared to other techniques as level set approach, topological derivative technique, and phase field method. The key part of density approach is the penalized interpolation function, which forces progressively porous cells made of fluid and solid simultaneously to belong discretely to fluid or solid sub-domains. However this type of problem was only solved in mono-fluid domains, in which the method accounts for the distribution of a single fluid and a single solid. The actual work aims to extend topology optimization in fluid flow problems to bi-fluid domain. A new interpolation function was developed for this purpose. Furthermore a penalization function was integrated in the multiobjective function, which ensure that each fluid takes its own path in the device, while maintaining a minimal required solid thickness between the channels of different fluids. The results showed the capacity of the proposed method to deal with multiple fluid phases in minimizing the pressure drop while maximizing heat exchange between different flows. The main conclusion is the potential of density approach to be applied on optimization of heat exchangers.

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1. Introduction

Optimization problems, in which the geometrical parameters are the optimization variables, are classified under three categories according to the degree of freedom, and the capability of changing the geometry: size, shape, and topology optimization. In size optimization, the problem variables are the geometrical parameters of the structure such as length, width, radius, etc. The shape and the connectivity of the elements between them are known and fixed. Hence, the final optimal solution is very similar to the initial baseline design. Shape optimization increases the degree of freedom of the problem, which can changes the size and the shape simultaneously by adding variables able to deform the boundaries of the structure. However, the architecture of the structure is still similar to the initial design. Finally, in topology optimization, every mesh in the optimization domain is a design parameter; this allows adding or removing material in every point in the design space without being limited to an initial topology. This increases significantly the number of variables in the optimization problem which make it more difficult to converge. However, topology optimization has become so appealing for its capacity to reach innovative and complex configurations based strictly on the defined objectives and constraints. After it was initially developed for mechanical structure problems, nowadays topology optimization is used in various

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Nomenclature

F	multiobjective function	μ	dynamic viscosity
G	constraint function	φ	porosity limit
k	thermal conductivity	ρ	density
N	number of design cells	σ	generic scalar for physical parameters
p_1	first penalization parameter	***Subscripts***	
p_2	second penalization parameter	i, j	cell indexes in horizontal and vertical directions
P	pressure		
Q	quantity	$f1$	fluid 1
Re	Reynolds number	$f2$	fluid 2
T	temperature	s	solid
u	velocity	i	inlet
w	weight of function	o	outlet
α	inverse permeability	c	continuity
η_1	first design parameter	d	fluid power dissipation
η_2	second design parameter	e	energy recovery
Γ	boundary		

engineering disciplines, including fluid mechanic problems. Mainly there are two big families in topology optimization techniques: discrete approaches and continuous approaches, which stand respectively for methods in which the local parameter controlling the material in each cell has discrete values, or continuous bounded values. These methods are gradient based optimization problems, thus the big number of design variables and the necessity to compute the derivative of the objective function with respect to the variables requires advanced mathematical techniques and software tools to ensure the convergence of the variables vector toward the optimal solution. Despite being widely used in physical problems such that mechanical structure and heat diffusion, discrete methods (ESO [1], BESO [2]) were not considered in mass transfer problems.

The application of topology optimization on fluid flow problems was first performed by Borrvall and Petersson [3] on Stoke flows using density method. Stokes flow is a type of fluid flow where the advective inertial forces are insignificant compared to viscous forces, thus having low Reynolds number ($Re \ll 1$). The optimization problem consists of minimizing the dissipated power in a fluid domain; the total volume of fluid should not exceed a maximum value considered as a constraint. They used density method in their work, which was the first application of topology optimization technique on fluid flow problems. The interpolation scheme introduced in [3] was widely used afterwards in topology optimization of mechanical fluid problems. Aage et al. [4] took advantage of parallel computations methods to solve the same problem of Borrvall and Petersson [3]. Gresborh-Hansen et al. [5] considered outflow rate as optimization target. Guest et al. [6] used Darcy's law of fluid flow in a porous medium, to deal with the presence of solid and fluid phases simultaneously in the optimization domain. The flow was then modeled by Darcy–Stokes equation. Wiker et al. [7] also treated Darcy–Stokes topology optimization problem. Their work included solution to an area to point drainage problem, where the goal was to transport all fluid out from the domain through a single part of the boundary with a minimal possible power consumption. They also studied the impact of many geometrical and mathematical parameters on the final solution. Contrary to [6], their results showed that regularization is needed in topology optimization of fluid problems to avoid numerical problems. Evgrafov et al. [8] stated that the problem of Darcy-Stokes flow in topology optimization admit solutions even if the limiting zero and infinite permeabilities of Darcy's law are included in the design domain. Gersborg-Hansen et al. [9] extended the application of topology optimization on Navier–Stokes flows for low and moderate Reynolds number. Deng et al. [10] used density method to solve topology optimization problem for unsteady flow. Results showed that the final optimum design is influenced by the dynamic effect and Reynolds number. The works of Kreissl et al. [11] on unsteady flow

showed that more the problem become unsteady, more the fluid channel architectures differ from those obtained in steady state problems. Deng et al. [12] solved the problem of topology optimization of flows driven by body forces. Their work included steady and unsteady flows. Results showed that in both types of flows, optimal design depends strongly on the type of body force.

Beside density method, level set technique gained attention in applying topology optimization on fluid flow problems. Zhou and Li [13] conducted 2D and 3D numerical experiments in topological design using level set method, and proved a relatively good agreement of their results with those obtained by density methods. Duan et al. [14] considered the bend pipe and diffuser cases in level set method, which were solved in density method by various authors [3, 9] for low Reynolds number. However, even if for some specific configurations level set method showed agreement with density method, the performance of this technique is still significantly limited by its incapacity of creating new holes in the design domain, which make the method highly dependent on the initial guess. Incorporating topological derivative method into level set method to reduce its dependency on the initial guess was considered in mechanical structure problems [15], then in mechanical fluid problems [16, 17]. Topological derivative indicates the location in the optimization domain where new holes should be nucleated using two different strategies: by including topological sensitivity information in the evolution equation of level set method, or using this information at discrete places in the optimization algorithm [17]. However results showed that the combined method of level set and topological derivatives still depend strongly on the starting guess, as discussed in [18].

In 2007, Bruns [19] applied topology optimization on convective heat transfer problems, by adding to heat transfer equation a convection term but without solving the fluid flow problem. Coupled heat and mass transfer in topology optimization were considered by Matsumori et al. [20] using the density approach. The single objective function consisted of maximizing heat transfer, whereas the input power that compensates the pressure drop is kept constant throughout the optimization. The optimization problem was formulated for two types of heat conditions: temperature dependent heat sources and temperature independent heat sources. A high level programing language and the commercial finite element software Femlab, were used by Okkels et al. [21] in the optimization of micro-cooling system. Multiobjective optimization by means of weighted sum approach was applied by Dede [22] on topology optimization of a heat and mass transfer problem. The two objectives functions were the mean temperature of the domain and the fluid power dissipation. The method was then applied, by the same author, on the optimization of the topology of an electronic device that generates heat to keep its maximum temperature below an acceptable limit [23], and to optimize an hierarchical microchannel cooling device [24]. In [25], Dede used two different interpolation schemes, one for the calculation of thermal conductivity and another one for the calculation of the inverse permeability using the same design variable. Yoon [26] minimized the thermal compliance in a structure subjected to a constant heat generation. In [27], Koga et al. developed a heat sink device using a multiobjective topology optimization technique. When a higher weight is given for heat transfer rate maximization at the detriment of minimization of pressure drop, small auxiliary channels appears in the domain.

The present work aims to extend the topology optimization methods in fluid mechanics to configuration that include two fluids flow, in addition to the solid phase that constitute the structure of the optimized device. The article is organized as follows: in Section 2, the use of density approach for topology optimization method is well explained, and the current optimization method and the physical equations that describe the problem are presented. In Section 3, the different computations techniques used in the optimization method are presented, then Section 4 is dedicated to the discussion of the results of the bi-fluid topology optimization method developed.

2. Topology optimization problem

2.1. Introduction

In topology optimization, the domain is discretized into design cells, which hold a design variable η that defines the material phase in the cell. For example, in mechanical structure problems, if $\eta = 1$ the cell is made of solid material, if $\eta = 0$ the cell is void. Hence, the values of all design variables in the discretized domain define the topology and the shape of the entire structure. In a similar manner, in fluid flow problems 0 means the cell is solid and 1 means the cell is fluid, which defines the fluid pipe architecture in the domain. Hence η can take two discrete values, 0 or 1. However, in order to avoid the difficulties related to solving the discrete problem, the design parameter is considered continuous in the interval [0,1], where the intermediate values]0,1[represent a porous media, which could be defined as a mix of void and materials in structure problems, or as a mix of fluid and solid in fluid problems. This topology optimization method is known as the density method (also known as SIMP method) which was developed by Bendsøe in 1989 [28] in mechanical structure problem to find the optimum shape of mechanical structures in order to maintain a certain load using a limited quantity of materials. The physical quantities in each cell are computed in function of the materials physical constants and the design variable η, using a penalized interpolation function. The key part of density method is the penalization parameter in the interpolation function that forces design variables to take discrete values (0 or 1) progressively throughout the optimization process, while ensuring a real physical representation of cells having intermediate densities at intermediate stages of the optimization process. SIMP was also largely used in heat conduction problem to find optimal distribution of materials having different thermal conductivities [29–31]. Figure 1 shows the initial "gray solution" where η have intermediate values in the domain and the convergence process to the final "black and white solution" having discrete values of η. The black and the white colors represent the materials which their topology is optimized ($\eta = 0,\ 1$) and the gray represent the fictitious intermediate material made of a mix of the black and the white ($\eta \in]\ 0, 1[$). Bi-fluid topology optimization method consists of optimizing material distribution of three phases in the domain, two fluids and one solid.

2.2. Problem formulation

The optimization domain Ω, is defined as the union of the following: solid subdomain Ω_s, fluid 1 subdomain Ω_{f1}, and fluid 2 subdomain Ω_{f2} ($\Omega = \Omega_s \cup \Omega_{f1} \cup \Omega_{f2}$). Boundary conditions could be:

- Walls with no slip assumption that could be either, at constant temperature T_w or adiabatic ($\partial T/\partial n = 0$).
- Fluid 1 inlet having a parabolic profile u_{i1} and a constant temperature T_{i1}.
- Fluid 1 outlet having a parabolic profile u_{02} and a null temperature gradient ($\partial T/\partial n = 0$).

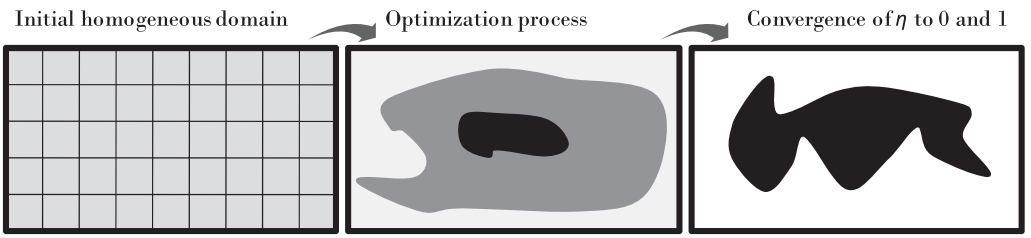


Figure 1. Optimization process in density method.

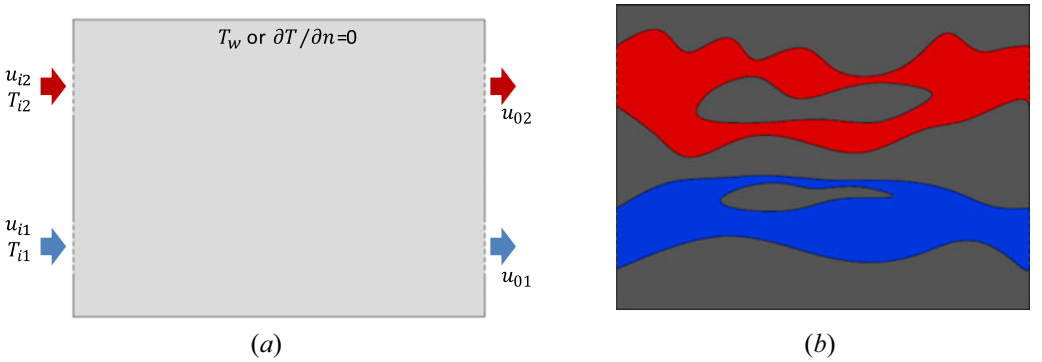


Figure 2. Initial domain (*a*) and final possible solution (*b*) in topology optimization of bi-fluid domain.

- Fluid 2 inlet having a parabolic profile u_{i2} and a constant temperature T_{i2}.
- Fluid 2 outlet having a parabolic profile u_{02} and a null temperature gradient ($\partial T/\partial n = 0$).

Both fluids are considered Newtonian and incompressible in a steady state laminar regime. Figure 2*a* shows a gray uniform initial domain and Figure 2*b* shows an example of how a final solution will look like. The initial domain is made of cells having a single intermediate value of the design variable η, which means each cell is made of a fictitious material which is a mix of fluid 1, fluid 2, and solid. The final domain is made of cells having discrete values of η which means each cell is strictly fluid 1, fluid 2, or solid.

2.3. Heat transfer and fluid flow modeling

Fluid flow and heat transfer phenomena are solved using the Navier–Stokes and energy equations expressed as follows:

$$\nabla.u = 0 \tag{1}$$

$$\rho(\eta)(u.\nabla u)u + \alpha(\eta)u = -\nabla P + \mu(\eta)\nabla^2 u \tag{2}$$

$$\left(\rho C_p(\eta)\right)u.\nabla T = \nabla.\left(k(\eta)\nabla T\right) \tag{3}$$

where u, P, and T are the velocity, the pressure, and the temperature, respectively, ρ, μ, and C_p are the density, the dynamic viscosity, and the specific heat capacity of the equivalent fluid, respectively. To allow modeling fluid, solid and intermediate phases using one momentum equation, inverse permeability coefficient α is added to Navier–Stokes equation similarly to Darcy's law that models fluid flow in a porous media, as suggested by Borrvall and Petersson in [3]. α varies from 0, for a pure fluid cell to a very large value for a pure solid cell, which forces the velocity to go to zero in these solid cells. For porous cells, α is computed in function of η using the penalized interpolation function ($\alpha_s > \alpha(\eta) > \alpha_{f1}$& α_{f2}). This slows down the velocity in these cells due to the presence of solid quantity that acts like a friction force to the fluid, similarly if there was a solid obstacle.

Theoretically in order to have zero velocity in solid regions, α should be $\alpha = \infty$ for solid and $\alpha = 0$ for fluid to model pure fluid flow without obstacles. In practice the two limits cannot be applied, so α should be bounded so that the lower value is $\alpha = \alpha \ll \mu$ and upper value is $\alpha = \overline{\alpha} \gg \mu$, as suggested in [3]. The inverse permeability value to enforce zero velocity in solid regions depends on Reynolds number. Kreissl et al. [11] studied the velocity field in solid regions for various values of α_s at different Reynolds number and showed that when Reynolds number decreases larger value of α_s are needed to force the velocity to go to zero in solid regions.

However results showed that $\alpha_s = 10^5$ is sufficient to ensure zero velocity in a solid cell for a wide range of Reynolds number.

2.4. Optimization problem

The optimal design parameter η is computed by minimizing or maximizing and objective function F. This allows defining an optimal geometrical shape and topology of solid, fluid 1 and fluid 2 subdomains in the optimization domain. The objective function depends on the flow velocity, pressure and temperature fields (computed in Eqs. (1)–(3)), denoted respectively u, P, and T. The physical quantities present in Eqs. (1)–(3) are evaluated by the interpolation function, which means that F depends also indirectly on the design parameter η. The optimization problem constraints $G_{f1}(\eta)$ and $G_{f2}(\eta)$ represent the maximum fraction of each fluid in the optimization domain, which will be equivalent to the maximum volume occupied by fluid channels in the final converged solution. The optimization problem is defined as follows:

$$\begin{aligned} \min_{\eta} \quad & F(z(\eta),\eta) \quad \text{with } z(\eta) = f(u,P,T) \\ \text{subject to} \quad & \text{Eqs. (1)–(3)} \\ & G_{f1}(\eta) \le \varphi_{f1} \\ & G_{f2}(\eta) \le \varphi_{f2} \end{aligned} \tag{4}$$

where φ_{f1} and φ_{f2} represent respectively the maximum global fractions of fluid 1 and fluid 2 in the optimization domain, hence $\varphi_{f1} + \varphi_{f2}$ will be equal to the maximum global porosity, to avoid filling entirely the domain with fluids, which is the best solution regarding pressure drop minimization.

2.5. Algorithmic scheme

The density approach discussed above consists of computing at each iteration an optimal value for the local design parameter η stored in each element over the discretized domain. The impact of each element's design variable η_i (i being the element index in the discretized domain) on the objective function is provided by the objective function's gradient and serves as the convenient information to the numerical optimization method to converge toward the optimal solution. It should be noted that density approach method consists of increasing progressively the penalization parameter p of the interpolation function, until reaching a value that allows the convergence to a black-and-white solution. Hence the optimization algorithm is divided into two loops:

- *Main loop* consists of increasing p progressively until it reaches the value that ensures global convergence to 0–1 solution, while avoiding fast convergence or to be stuck in a local minimum. This consists of finding the optimal solution η^1_{opt} for a penalization parameter p_{v1} in the interpolation function, then to use this optimal solution, η^1_{opt}, as an initial solution for the same problem with a new value for the penalization parameter denoted p_{v2}, such as $p_{v2} > p_{v1}$. This procedure is repeated until reaching a domain made of discrete values (0 or 1) of the design parameter η in all cells. (For further details please refer to [3])

  Inner loop (Figure 3) which consists of the following steps until convergence:

1. Calculation of cell physical properties using the penalized interpolation function and η values from the previous iteration.
2. Solving fluid flow equations by means of finite volume method.
3. Evaluation of the objective function F.

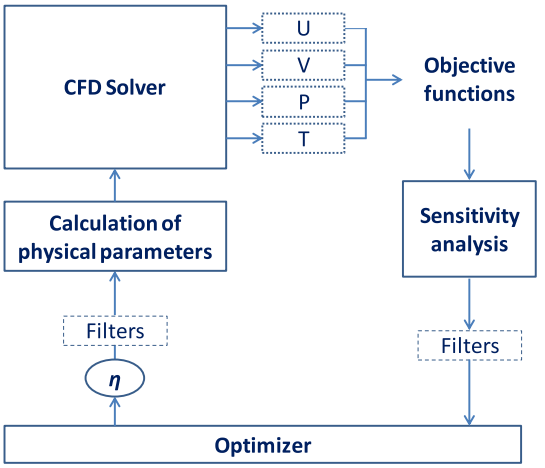


Figure 3. Inner loop.

4. Calculations of the gradient of the objective function with respect to η by means of discrete adjoint method.
5. Solving the numerical optimization problem using the method of moving asymptotes.

Several regularization functions (filters) are used at different level of the algorithm (applied on design and sensitivity fields). These regularization functions allow better convergence and avoid checkerboard problems [32].

3. Optimization method

3.1. Interpolation scheme

In [3], Borrvall and Peterson introduced the first interpolation scheme applied to topology optimization of fluid flow problems. They also provided the mathematical proofs that this scheme ensures the convergence to an optimal solution. Many authors have later adopted this interpolation scheme in topology optimization of fluid flow problems. The physical properties of the material in each cell in the discretized domain is computed in function of the design variable η and fluid and solid properties, according to the convex interpolation function defined as follows [3]:

$$\sigma(\eta, p) = \sigma_s + (\sigma_f - \sigma_s)\eta\frac{1+p}{\eta+p} \quad \text{with } 0 \le \eta \le 1 \text{ and } p > 0 \tag{5}$$

where σ represent the physical properties (for example inverse permeability coefficient) with σ_s and σ_f being respectively the values of this property for solid and fluid phases. The penalty parameter p defines the convexity of the function and has a crucial role in the convergence of the problem to a discrete solution. When the penalization parameter tends to zero, the penalization function become more convex and the optimization algorithm allows higher number of gray cells. At high values of penalization parameter ($p \approx 1$), the interpolation function will have pseudo linear shape and will reduce the number of gray cells. The level of gray material in function of the penalization parameter p, is well studied and demonstrated in [3]. Borrvall and Peterson suggested starting the optimization with low penalty values and gradually increasing the penalization parameter, as already discussed in Section 2.5.

The interpolation scheme presented above is available for mono-fluid topology optimization problem only. Therefore; a new interpolation function suitable for bi-fluid problems is formulated

in this work, on the basis of multi-material interpolation scheme largely used in topology optimization of mechanical structure problems. A multi material interpolation scheme based on SIMP function was developed by Sigmund [33] for three-phase optimization problem, two materials and void. This interpolation scheme was also used by Zhuang et al. [34] in the topology optimization of transient heat conduction problems having 3 phases, two materials and a void phase. The method consisted of interpolating first between the two materials, than to interpolate between the equivalent material the void phase. This means that the final interpolation scheme will be consisted of two interpolation functions merged together in a single function. This interpolation model appears to be convenient for bi-fluid topology optimization problems, in which fluid 1 and fluid 2 will form together an equivalent fluid, which in his turn form with the solid phase a single homogeneous porous media that has a single velocity vector. The general form of multimaterial interpolation function is the following:

$$\sigma(\eta_1,\eta_2) = g(\eta_1,p_1)\left[g(\eta_2,p_2)\sigma_A - \left(1-g(\eta_2,p_2)\right)\sigma_B\right] + \left(1-g(\eta_1,p_1)\right)\sigma_C \tag{6}$$

In case of bi-fluid topology optimization problem, following the interpolation logic explained above we will have: $\sigma_A = \sigma_{f1}$, $\sigma_B = \sigma_{f2}$, and $\sigma_C = \sigma_s$, where σ_{f1}, σ_{f2}, and σ_s correspond respectively to fluid 1, fluid 2, and solid physical quantities. Equation (6) shows that η_1 corresponds to the fraction of the equivalent fluid (fluid 1 + fluid 2), and η_2 is the fraction of fluid 2 in η_1 fraction. Finally replacing $g(\eta,p)$ by its expression developed in [3] for mono-fluid topology optimization problem $\left(g(\eta) = \eta(1+p)/(\eta+p)\right)$. Eq. (6) is then expressed as follows:

$$\sigma(\eta_1,\eta_2) = \eta_1\frac{1+p_1}{\eta_1+p_1}\left[\eta_2\frac{1+p_2}{\eta_2+p_2}\sigma_{f1} - \left(1-\eta_2\frac{1+p_2}{\eta_2+p_2}\right)\sigma_{f2}\right] + \left(1-\eta_1\frac{1+p_1}{\eta_1+p_1}\right)\sigma_s \tag{7}$$

Equation (7) is sketched in Figure 4a for $\sigma_{f1} = \sigma_{f2} \neq \sigma_s$ and in Figure 4b for $\sigma_{f1} \neq \sigma_{f2} \neq \sigma_s$. η_1 field defines the shape of fluid channels (solid cell for $\eta_1 = 0$ and equivalent fluid cell for $\eta_1 = 1$) and η_2 determines the type of fluid inside each cell of the channel (fluid 1 cell for $\eta_2 = 0$ and fluid 2 cell for $\eta_2 = 1$). The local design parameter η is now defined as follows:

$$\eta(x) = (\eta_1(x),\eta_2(x)) = \begin{cases} (0,\forall\eta_2(x)) \text{ if } x \in \Omega_s \text{ (solid cell)} \\ (1,0) \text{ if } x \in \Omega_{f1} \text{ (fluid 1 cell)} \\ (1,1) \text{ if } x \in \Omega_{f2} \text{ (fluid 2 cell)} \end{cases} \tag{8}$$

Finally fluid 1 and fluid 2 global fractions constraints are expressed as follows in function of η_1 and η_2 :

$$G_{f1}(\eta_1,\eta_2) = \frac{1}{N}\int_\Omega \eta_1(1-\eta_2)d\Omega \leq \varphi_{f1} \tag{9}$$

$$G_{f2}(\eta_1,\eta_2) = \frac{1}{N}\int_\Omega \eta_1\eta_2 d\Omega \leq \varphi_{f2} \tag{10}$$

N being the total number of meshes in the discretized domain.

3.2. Finite volume method

The finite volume method as introduced by Versteeg and Malalasekera [35] was considered in this work for the resolution of the physical equations presented in Section 2.3, taking into account the modifications introduced by Marck et al [36] to take into consideration the inverse permeability coefficient in Eq. (2). The authors applied the modified finite volume method on topology optimization of multi objective heat and mass transfer problem in a mono-fluid configuration [36]. They also introduced a shear force correction term that depends on the design

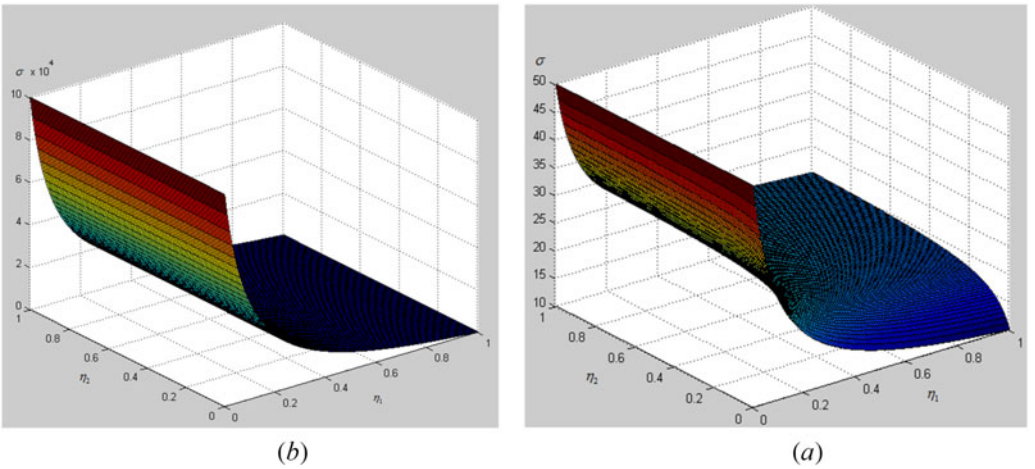


Figure 4. Bi-eta interpolation function.

variable η to treat the wall shear stress along the interface between fluid and solid cells, which was also used in this work.

The transport equations are discretized using finite volume method over a uniform cartesian grid, which is the same grid used for design variables. Only the velocity components are evaluated on staggered grids, where the control volume of the horizontal component of the velocity is centered on the east face of the design control volume and the vertical component is centered on the north face of the design control volume. The reason to use staggered grid for velocity is to avoid the "checker-board" problem defined by the nonphysical behavior when computing the pressure field if velocities are evaluated at the same control volumes as pressure, as discussed in [35]. The quadratic upwind differencing scheme (QUICK) was selected to calculate the coefficients of the discretized equations. This scheme was chosen because it takes into account the flow direction of the quantity transported, which is very important for the accuracy and the stability of the method specially for high Reynold number flows. Finally SIMPLER iterative algorithm is chosen to solve the coupled velocity-pressure problem seen in Eqs. (1) and (2) (for further details on SIMPLER refer to [35])

Under relaxation factor is implemented in the modified finite volume method in this work, in order to improve the convergence of the resolution of the system of equations $AX = B$, with X being the velocity, pressure or temperature fields. This factor allows to take into consideration X computed in the previous iteration of SIMPLER iterative procedure. Equation system becomes expressed as follows [37]:

$$\frac{A_i X_i}{\lambda_{\mathrm{UR}}} = \sum\nolimits_{\mathrm{nb}} A_{\mathrm{nb}} X_{\mathrm{nb}} + B + \frac{1-\lambda_{\mathrm{UR}}}{\lambda_{\mathrm{UR}}} A_i X_{i,\ \mathrm{old}} \tag{11}$$

where λ_{UR} is the under relaxation factor that varies between zero and one, and i and nb subscripts correspond respectively to a control volume in the discretized domain and its four neighbors control volumes at its east, west, north, and south.

3.3. Objective functions

The optimization problem is now defined as follows: find the optimal values of design variables that define the optimal distribution of three phases, solid fluid 1 and fluid 2 (Figure 2), in order to minimize fluid power dissipation in the domain and maximize heat transfer, without exceeding the maximum allowable fraction of each fluid. The objective functions of pressure drop and heat transfer are defined respectively as follows:

$$f_d(u,P) = \int_{\Gamma_{out}} u\left(P + \frac{1}{2}\rho u^2\right)d\Gamma \; - \; \int_{\Gamma_{in}} u\left(P + \frac{1}{2}\rho u^2\right)d\Gamma \tag{12}$$

$$f_e(u,T) = \int_{\Gamma_1} \mathrm{n.u}\left(\rho C_p T\right)d\Gamma \; - \; \int_{\Gamma_2} \mathrm{n.u}\left(\rho C_p T\right)d\Gamma \tag{13}$$

where Γ_{out} and Γ_{in} denote respectively for outlet and inlet boundaries of both fluids. Γ_1 denotes for the inlet and outlet boundaries of fluid 1, and Γ_2 denotes for the inlet and outlet boundaries of fluid 2. Fluid 1 and 2 are considered the cold fluid and hot fluid respectively. n is a vector normal to the surface to have always a positive value for $n.u$ on outlet boundary and a negative value on the inlet. f_e is equal to the double of heat transfer quantity, because it is computed on the boundaries of both fluids for numerical reasons regarding sensitivity analysis. The objective functions f_d and f_e are coupled in a single multiobjective function, using weighted sum method. It should be noted that an increasing in f_e leads to an increasing in f_d and vice versa for optimal solutions having different geometries. Hence the tradeoff solution between those two functions is a Pareto optimal solution. In order to have the same order of magnitude, and to let the weight reflects the preference of each objective and not to compensate the difference in the magnitudes, f_d and f_e are normalized, and the multiobjective function is expressed as follows:

$$F = w\frac{f_d - f_d^{\min}}{f_d^{\max} - f_d^{\min}} \; - \; (1-w)\frac{f_e - f_e^{\min}}{f_e^{\max} - f_e^{\min}} \tag{14}$$

$f_d^{\min}$ and $f_e^{\min}$ are the results of the optimization problem solved with $w = 1$, and $f_d^{\max}$ and $f_e^{\max}$ are the results of the optimization problem solved with $w = 0$.

Design and optimization of devices that include two flow of two different fluids requires that each fluid has his own flow channel independently from the second fluid. On the other hand the inverse permeability coefficient of fluid 1 and fluid 2 are both equal to zero $(\alpha(\eta_1 = 1) = 0)$. Thie means that the presence of fluid 1, fluid 2 or any mixture of them will have no effect on velocity and pressure fields, thus no effect on minimizing fluid power dissipation (the gradient of power dissipation objective function with respect to η_2 will be null). Therefore, an objective function, called continuity function, that ensure the continuity of the flow of each fluid between its predefined inlet and outlet boundaries, is added to the multiobjective function in order to ensure the separation of the channels formed by different fluids subdomains. The continuity function is expressed as follows

$$f_c = \frac{1}{N}\sum_{i,j}^{N_x,N_y}\frac{1}{4}Q_{f1}^{i,j}\left[Q_{f2}^{i-1,j} + Q_{f2}^{i+1,j} + Q_{f2}^{i,j+1} + Q_{f2}^{i,j-1}\right] + \frac{1}{4}Q_{f2}^{i,j}\left[Q_{f1}^{i-1,j} + Q_{f1}^{i+1,j} + Q_{f1}^{i,j+1} + Q_{f1}^{i,j-1}\right] \tag{15}$$

where $Q_{f1}^{i,j} = \eta_1^{i,j}\left(1 - \eta_2^{i,j}\right)$ and $Q_{f2}^{i,j} = \eta_1^{i,j}\eta_2^{i,j}$ are respectively the quantities of fluid 1 and fluid 2 in a cell (i,j) and N is the number of meshes in the domain (number of local design variables). Continuity objective function has values greater than 0 when the cell (i,j) and its neighbors are made from different fluids; hence the objective function will only prevent two adjacent cells from containing two different fluids. It should be noted also that Eq. (15) prevents also that two different fluids exist simultaneously in the same cell when the algorithm converges to the final optimal solution. The multiobjective function (14) is now expressed as follows:

$$F = w\frac{f_d - f_d^{\min}}{f_d^{\max} - f_d^{\min}} \; - \; (1-w)\frac{f_e - f_e^{\min}}{f_e^{\max} - f_e^{\min}} + w_c f_c \tag{16}$$

where w_c is a numerical parameter to adjust the weight of the continuity equation in the multiobjective function which ensure the smooth convergence of the problem. f_c could be considered as

penalty function, hence it converges to 0 in the final solution where no fluid 1 or fluid 2 are present in the same cell or two neighbors cells.

3.4. Discrete adjoint method

The next step in topology optimization algorithm requires the evaluation of the objective function's derivative with respect to the design variables. A widely used technique in shape and topology optimization for computing the sensitivity of the objective function is the adjoint method, in its two forms: continuous form and discrete form. The difference between continuous and discrete adjoint is the following: in continuous adjoint, all the equations are derived analytically from their initial forms (Eqs. (1)–(3)), and then discretize. In discrete adjoint method, the discretized forms of physical equations are derived. According to Nadarajah and Jameson [38], the discrete gradients have better agreement with finite difference gradients than the continuous gradients. However it was shown that this difference is small and the difference between the three methods is reduced when the mesh size decreases. In this work discrete adjoint approach is adopted for the calculation of the gradient.

Consider an objective function $f(X(\eta))$ subject to $J(X,\eta) = AX - B = 0$, the residual of the resolution of the system of equations in finite volume method with $X = U,\ V, P$, or T ($U,\ V, P$, and T being respectively the horionzontal velocity component, the vertical velocity component, the pressure and the temperature). The Lagrange function is introduced and defined as follows [39]:

$$L(X,\eta) = f(X) + \lambda^t . J(X,\eta) \tag{17}$$

where λ^t is the adjoint vector, with the same dimension as the X vector. When X satisfies J, the Lagrange function L is equal to the objective function f. The total derivative of Eq. (17) with respect to η is:

$$\frac{dL}{d\eta} = \frac{df}{d\eta} + \lambda^t \frac{dJ}{d\eta} \tag{18}$$

Equation (18) could be written as follows in function of partial derivatives of the objective and residual functions [39]:

$$\frac{dL}{d\eta} = \lambda^t \frac{\partial J}{\partial \eta} + \frac{\partial f}{\partial \eta} + \left[\frac{\partial f}{\partial X} + \lambda^t \frac{\partial J}{\partial X}\right] \frac{\partial X}{\partial \eta} \tag{19}$$

The critical term to evaluate in Eq. (19) is $\partial X/\partial \eta$. Discrete adjoint approach suggests that the adjoint vector verifies the equation multiplied by $\partial X/\partial \eta$, hence it does not need to be evaluated. The evaluation of $df/d\eta$ could be hence computed by solving the following systems of equations:

$$\begin{cases} \left(\frac{\partial J}{\partial X}\right)^t \lambda = -\left(\frac{\partial f}{\partial X}\right)^t \\ \frac{df}{d\eta} = \lambda^t \frac{\partial J}{\partial \eta} + \frac{\partial f}{\partial \eta} \end{cases} \tag{20}$$

The term $\partial f/\partial \eta$ is null for f_d and f_e, whereas the term $\partial f/\partial X$ is null for f_c. The adjoint vector λ and residual function J have the following forms:

$$\lambda = \begin{pmatrix} \lambda_U \\ \lambda_V \\ \lambda_P \\ \lambda_T \end{pmatrix} \quad J = \begin{pmatrix} AX_U - B_U \\ AX_V - B_V \\ AX_P - B_P \\ AX_T - B_T \end{pmatrix} \tag{21}$$

The matrices $\partial J/\partial X$ and $\partial J/\partial \eta$ are expressed as follows:

$$\frac{\partial J_X}{\partial \gamma} = \underbrace{\begin{pmatrix} \cdots & \overbrace{\begin{matrix}\vdots\\ \vdots\\ \vdots\end{matrix}}^{\frac{\partial A_X}{\partial \gamma_{i,j}}.X - \frac{\partial B_X}{\partial \gamma_{i,j}}} & \cdots \\ \cdots & & \cdots \\ \cdots & & \cdots \end{pmatrix}}_{=N_\gamma} \Bigg\}_{=N_X} + \begin{cases} A_X \ if \ X = \gamma \\ 0 \ if \ X \neq \gamma \end{cases}$$

Figure 5. Matrix form of the derivative of the residual function with respect to physical and design variables.

$$\frac{\partial J}{\partial X} = \begin{pmatrix} \frac{\partial J_U}{\partial U} & \frac{\partial J_U}{\partial V} & \frac{\partial J_U}{\partial P} & 0 \\ \frac{\partial J_V}{\partial U} & \frac{\partial J_V}{\partial V} & \frac{\partial J_V}{\partial P} & 0 \\ \frac{\partial J_P}{\partial U} & \frac{\partial J_P}{\partial V} & \frac{\partial J_P}{\partial P} & 0 \\ \frac{\partial J_T}{\partial U} & \frac{\partial J_T}{\partial V} & 0 & \frac{\partial J_T}{\partial T} \end{pmatrix} \quad \frac{\partial J}{\partial \eta} = \begin{pmatrix} \frac{\partial J_U}{\partial \eta_1} & \frac{\partial J_U}{\partial \eta_2} \\ \frac{\partial J_V}{\partial \eta_1} & \frac{\partial J_V}{\partial \eta_2} \\ \frac{\partial J_P}{\partial \eta_1} & \frac{\partial J_P}{\partial \eta_2} \\ \frac{\partial J_T}{\partial \eta_1} & \frac{\partial J_T}{\partial \eta_2} \end{pmatrix} \tag{22}$$

The assembly of $\partial J/\partial X$, and $\partial J/\partial \eta$ lies on the derivatives of $J_X = A_X.X - B_X$ with respect to every term of a vector γ with $\gamma = X$ or η. The derivative of J_X with respect to γ is expressed as follows:

$$\frac{\partial J_X}{\partial \gamma} = \frac{\partial A_X}{\partial \gamma} X - \frac{\partial B_X}{\partial \gamma} + A_X \frac{\partial X}{\partial \gamma} \tag{23}$$

Equation (23) could be expressed also in matrix form as seen in Figure 5. $N_X = N_x(X) \times N_y(X)$ and $N_\gamma = N_x(\gamma) \times N_y(\gamma)$, N_x and N_y being the number of meshes in horizontal and vertical directions respectively in the discretized domain for X and γ fields, and i and j denotes respectively the horizontal and vertical indexes of a control volume in the domain.

3.5. Method of moving asymptotes

The Method of Moving Asymptotes (MMA), adopted in this work, was developed by Svanberg in 1987. The method is based on a special type of convex approximation [40, 41]. It can fit single and multiple constraints, and proved to be efficient in topology optimization problems. MMA has become a standard optimization tool for topology optimization problems, especially with density based methods [42]. Svanberg proposed a better convergent version of MMA in 2002, the globally convergent version of the method of moving asymptotes (GCMMA) [43]. However GCMMA needs relatively higher calculation time, hence it was less used than the standard MMA in the scientific literature.

MMA solves the optimization problem defined as follows:

$$\begin{aligned} &\text{minimize } f_0^k(\eta) \\ &\text{subject to } f_i^k(\eta) - b_i \leq 0 \quad i = 1, \ldots, m \ \ \& \ \ \eta_{min} < \eta < \eta_{max} \end{aligned} \tag{24}$$

where m is the number of constraints, n the number of variables and k the index of the iteration in the optimization loop. At each iteration in the optimization procedure a subproblem is created, in which the functions $f(\eta)$ (objective function and/or constraint functions) are replaced by an approximated convex function $\tilde{f}^k(\eta)$ [41]. $\tilde{f}^k(\eta)$ is computed in function of the gradient of f with

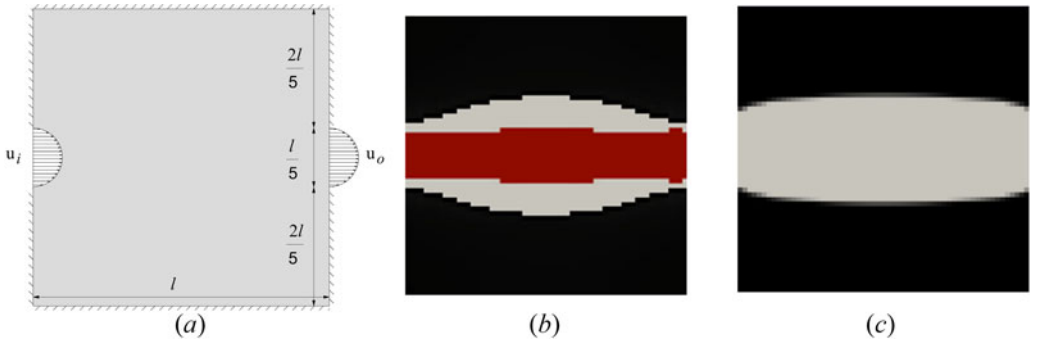


Figure 6. Initial configuration (*a*) and topology optimization results for case 1 (*b*) and case 2 (*c*).

respect to η^k and the moving asymptotes at iteration k which are computed in function of η^k, η^{k-1} and the moving asymptotes of iteration $k-1$. The asymptotes that limit the search space of the optimal point are not fixed but updated at each iteration, therefore the method is called MMA. Many authors proposed new methods to update the approximation functions [42, 44, 45], and to improve the convergence of the original method proposed by Svanberg [41]. The subproblem generated by MMA is then solved using "primal-dual-interior-point method", as presented in [41], to find the optimal point η^{k+1} which is considered the initial solution of the next iteration.

4. Topology optimization results

The first part of the results focuses on testing the capacity of the bi-fluid topology optimization method described in this article on responding to an optimal positioning of three different materials in the optimization domain, two fluids and one solid, in heat and mass transfer problems. Hence, in this part the penalty continuity function f_c is omitted from the mutliobjective function ($w_c = 0$). Three cases were studied in this part, first one by considering different inverse permeability coefficients for the three materials ($\alpha_{f1} \neq \alpha_{f2} \neq \alpha_s$), the second one by considering different maximum fluid porosities ($\varphi_{f1} \neq \varphi_{f2}$) and finally a third case with different thermal conductivities ($k_{f1} \neq k_{f2} \neq k_s$). The second part of the results aims to optimize the shape of fluid channels by minimizing pressure drop due to fluid flow and maximizing heat transfer between a fluid 1 considered the cold fluid and a fluid 2 considered the hot fluid. The creation of different channels for each fluid alone separated by a solid domain is ensured by the penalty continuity function f_c ($w_c \neq 0$).

4.1. Minimization of pressure drop and maximization of heat transfer without channels separation (w_c=0)

4.1.1. Mass transfer ($w = 1$)

The design domain in Figure 6a is square shaped of 60 mm side length having a single inlet and outlet sections of fluid. Both fluids are considered having the same density and dynamic viscosity ($\rho = 1000$ kg/m^3, $\mu = 0.001$ Pa/s). Inlet and outlet fluid boundaries are subjected to a parabolic velocity profile. The Reynolds number of the fluid at inlet, based on the average inlet velocity and its inlet dimension $l/5$, is $Re = 12$. The remaining of the boundaries are adiabatic walls with no-slip conditions. In this paragraph only minimization of pressure drop is considered in the optimization problem ($w = 1$). Two cases are studied in the current configuration:

Case 1: $\varphi_{f1} = \varphi_{f2} = 0.18$, and $\alpha_{f1} = 10^3$, $\alpha_{f2} = 0$ and $\alpha_s = 10^7$. This correspond in having three materials in the optimization problem. A fluid 2 without an inverse permeability coefficient, a solid having a very large inverse permeability coefficient hence it is impossible to have a fluid motion in the domain of this material, and a fluid 1 having an intermediate value of the inverse

permeability, hence the flow in fluid 1 domain will be slowed down similarly if there was a solid obstacle (equivalent to a flow through a porous media).

Case 2: $\varphi_{f1} = 0.1$, $\varphi_{f2} = 0.26$, $\alpha_s = 10^7$ and $\alpha_{f1} = \alpha_{f2} = 0$. This correspond in having a pure solid domain with no possible fluid flow in it, and two pure fluid domains (fluid 1 and fluid 2). Both fluid differ only by the maximum allowed fluid 1 and fluid 2 fractions, which will have an impact only on η_2 field.

Figure 6*b* shows the topology optimization result for case 1. The black color represents the solid domain, defined by $\eta_1 = 0$, the red color represents fluid 2 domain defined by $\eta_1 = 1$ & $\eta_2 = 1$, and the white color represents the fluid 1 domain defined by $\eta_1 = 1$ & $\eta_2 = 0$. Fluid 2 which has zero inverse permeability coefficient, is accumulated in the middle of the channel and fluid 1 which has $\alpha_{f1} > \alpha_{f2}$ is accumulated in the remaining space between fluid 2 and solid domains. In order to clarify the effect of this optimization result and domains distribution on velocity and pressure fields, hence on the fluid power dissipation objective function $F = f_d$, the latter is computed for the following cases:.

- The optimal case represented in Figure 6b: $f_d^1 = 2.19 \times 10^{-8}$ W.
- The same configuration of the optimal result case but inverting fluids inverse permeabilities ($\alpha_{f1} = 0$ and $\alpha_{f2} = 10^3$): $f_d^2 = 1.02 \times 10^{-7}$ W.
- Considering a constant value $\eta_2 = 0.5$ in all the optimization domain, which correspond to replacing fluid 1 and fluid 2 domains by a single equivalent fluid domain having α interpolated between $\alpha_{f1} = 10^3$ and $\alpha_{f2} = 0$ with $\eta_2 = 0.5$ and $p_1 = p_2 = 1$ using Eq. (7) (η_1 remaining the same because no changing in the topology of the solid domain): $f_d^3 = 2.21 \times 10^{-7}$ W

$f_d^1 < f_d^2 < f_d^3$ which explains the conversion of the optimization method to such a configuration by preferring to accumulate fluid 2 in the center of the domain.

Figure 6*c* shows η_1 field for the optimal result of case 2. Black color represents $\eta_1 = 0$, hence solid domain Ω_s, and white color represents $\eta_1 = 1$ hence fluid 1 and fluid 2 domains $\Omega_{f1} \cup \Omega_{f2}$. To take advantage of the $\varphi_{f2} - \varphi_{f1} = 0.16$ excess volume of fluid 2 in comparison with fluid 1, η_2 converges to values between 0.71 and 0.74 in all fluid cells that have $\eta_1 \approx 1$. This means that in average each cell is made of 72.5% of fluid 2 and 27.5% of fluid 1, which is approximately equal to X, the ratio of fluid 2 mean porosity over the calculation domain, $X = \varphi_{f2}/(\varphi_{f2} + \varphi_{f1}) = 0.26/(0.26 + 0.1) = 0.722$. These values of η_2 in non-solid cells implies that the sum of both constraints functions of the problem $G_{f1} + G_{f2}$ are approximately equal to total fluid maximum fractions $\varphi_{f1} + \varphi_{f2}$. This allowed to use the entire maximum quantity of fluid (fluid 1 + fluid 2) in the optimization problem to converge to the topology that corresponds to the minimum of the fluid power dissipation f_d.

4.1.2. Heat and mass transfer ($w < 1$)

The example studied in the previous paragraph is reconsidered with minimizing pressure drop and maximizing heat transfer simultaneously in a single multi-objective optimization problem ($w < 1$ and $\Gamma_2 = \text{Null}$). The west side is subjected to an inlet boundary condition with a parabolic velocity profile and an inlet fluid temperature $T_i = 0$ °C, whereas the outlet section is subjected to a similar velocity profile with a null temperature gradient. The entire south and north boundaries are subjected to a fixed temperature $T_w = 10$ °C as seen in Figure 7*a*. The rest of the west and east boundaries are adiabatic walls with no slip condition. The physical properties are the following:$\rho = 1000$ kg/m^3, $\mu = 0.001$ Pa/s, $C_p = 5$kj/K kg, (ρ, μ, and C_p are similar for fluid 1 and fluid 2) $\alpha_{f1} = 0$, $\alpha_{f2} = 0$, $\alpha_s = 10^7$, $k_s = 10$ W/kg m, $k_{f1} = 0.01$ W/kgm and $k_{f2} = 1$ W/kgm. A single constraint function is considered in the current $G_s(\eta_1) = \frac{1}{N}\int_\Omega 1 - \eta_1 \, d\Omega \geq \varphi_s = 0.6$, which represent the minimal fraction of total solid in the domain.

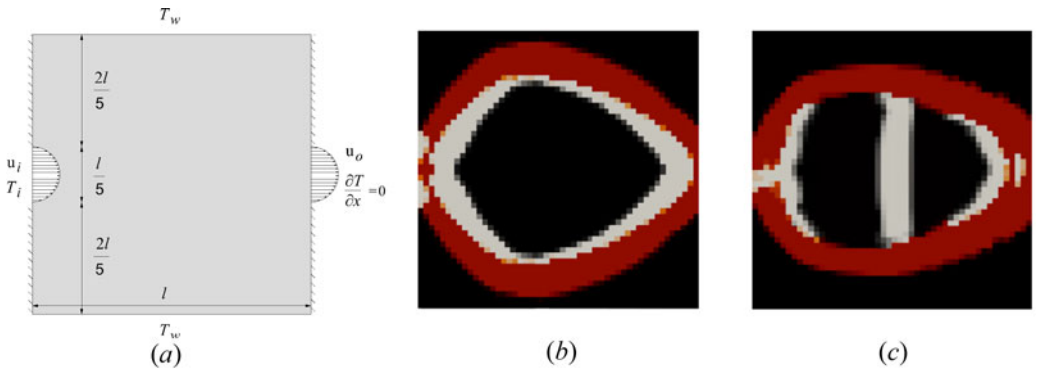


Figure 7. Initial configuration (*a*) and topology optimization results for case 3 with $w = 0.8$ (*b*) and with $w = 0.7$ (*c*).

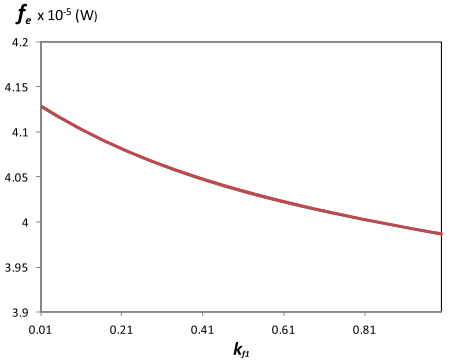


Figure 8. Variation of f_e in function of k_{f1} in optimal result configuration of Figure 7*a*.

Figures 7*b*, *c* show respectively the topology optimization result for $w = 0.8$ and $w = 0.7$. The topology optimization results in Figure 7*b* shows two main characteristics. First the single pipe is divided into two smaller pipes that approaches of north and south walls, which are the source of heat, to enhance thermal energy recuperation by the fluid. This change in channels geometries provokes an increasing in pressure drop, which reflect the opposition between minimization of pressure drop and maximization of heat transfer in the optimization of heat and mass transfer problems. This behavior is in coherence with the works of Marck et al. [36] on topology optimization of heat and mass transfer in mono-fluid domain, who considered a similar configuration. The second important feature that can be seen in the same result, is that both fluid 1 and fluid 2 are present in the non-solid domain ($G_{f1} = 0.142$ and $G_{f2} = 0.257$). These regions are formed by either fluid 1 or fluid 2 cells rather than a mix of them (convergence of η_2 to a 0-1 solution). In fact, fluid 1cells which have a thermal conductivity 100 times lower than fluid 2, are placed near the interface between flow regions and solid central core to thermally isolate it. This means less energy is dissipated by the fluid to increase the temperature of the central core; which increases f_e. Figure 8 highlights the importance of this distribution of Ω_{f1} and Ω_{f2} on heat recuperation objective function. In fact when k_{f1} increases from 0.01 to $k_{f1} = k_{f2} = 1$, heat recuperation decreases from $f_e = 4.13 \times 10^{-5}\ W$ to $f_e = 4.01 \times 10^{-5}\ W$.

Figure 7*c* shows topology optimization result for the same problem with $w = 0.7$. In this case, the algorithm used the low conductive fluid 1 to split the central core into two parts. In fact, an horizontal heat flux in the central core, takes place from near fluid outlet region to near fluid inlet region. Hence the outlet flow is cooled by conduction through the high conductive solid material in the central core, which decreases energy recovery f_e. Figure 9*a* shows that there is no velocity in the vertical fluid barrier in the central core. This means that this quantity of fluid used has no effect on

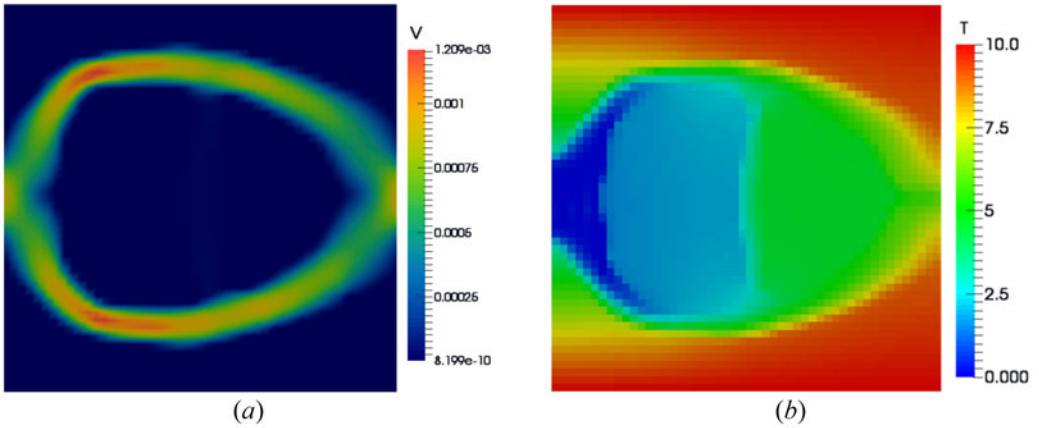


Figure 9. Velocity field (*a*) and temperature field (*b*) for optimal geometry of $w = 0.7$.

decreasing fluid power dissipation, and has only effect on minimizing heat diffusion through the solid core due to its low thermal conductivity. The fluid power dissipation function in this case is $f_d = 5.05 \times 10^{-8}$ W, which is higher by 48%, in relation to fluid power dissipation of the optimal result of the case of $w = 0.8$. However heat recovery is $f_e = 4.43 \times 10^{-5}$ W, which means an increment by 25.9% in relation to the previous case. Figure 9*b* highlight the temperature gradient between the parts of the central core at the right and the left of the low-conductive fluid 1 barrier, which shows clearly the effect on minimizing the horizontal heat flux.

4.2. Minimization of pressure drop and maximization of heat transfer with channels separation ($w_c > 0$)

4.2.1. Mass transfer ($w = 1$)

The rectangular optimization domain of length L and width l sketched in Figure 10, has an aspect ratio of 1.5 ($L = 1.5 \times l$). The domain is discretized into 72 design cells in horizontal direction and 48 design cells in vertical direction. Inlet and outlet sections of fluid 1 are located in the lower part of the domain (near south wall), whereas those of fluid 2 are located in its upper part (near north wall). Inlet and outlet boundaries of both fluids have similar parabolic profiles with Reynolds number equal to $Re = 48$, and the rest of the boundaries are adiabatic walls with no-slip condition. Maximum fractions of both fluids are $\varphi_{f1} = 1/6$ and $\varphi_{f2} = 1/6$, which correspond to the surface occupied by a straight pipe between inlet and outlet sections of each fluid. The optimization problem is solved in parallel flow configuration (Figure 10*a*) and in counter flow configuration (Figure 10*b*). At first, the optimization problem is solved without considering continuity function for the separation of the channels of both fluids, which means the problem is similar to a mono fluid topology optimization problem ($w_c = 0$). Topology optimization results are presented in Figure 11*a* for parallel flow and Figure 11*b* for counter flow. Without continuity function, the algorithm makes no difference between the inlet and outlet sections of fluid 1 and 2; and $\eta_2 = 0.5$ in the entire domain, which means all non-solid domain are composed from a single equivalent fluid, which is represented in orange color. The optimal structure of Figure 11*a* is similar to the one presented by the authors who solved the same configuration problem in mono fluid topology optimization [3, 6, 36, 46]. For this case the two pipes are joined in a single wider pipe in the middle of the domain and then disjoined near outlet region in order to get the lower pressure drop ($f_d = 8.6 \times 10^{-8}$ W). Borrval and Petersson [3] made a more extensive study on this configuration and stated that $L = 1.08 \times l$ is the critical aspect ratio, where for geometries having $L/l > 1.08$, joining pipes to minimize the effect of fluid-solid interface on

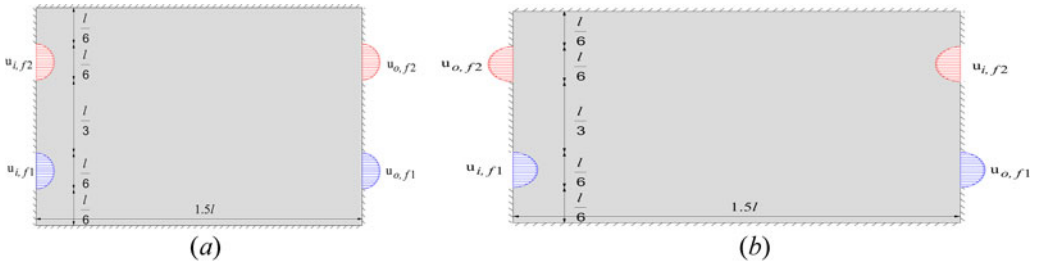


Figure 10. Initial configuration of double pipe example with AR = 1.5: (*a*) Parallel flow and (*b*) counter flow.

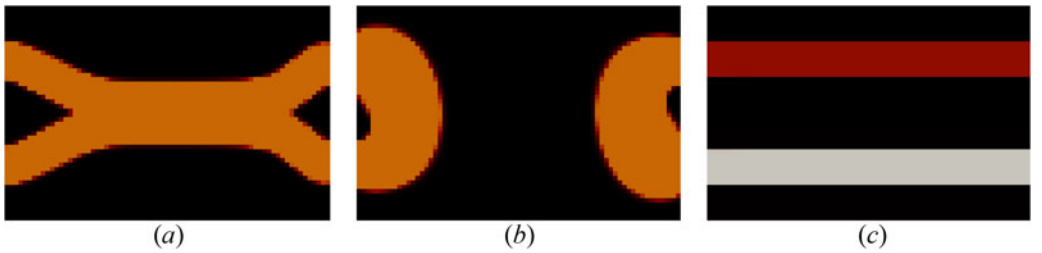


Figure 11. Topology optimization results: (*a*) Parallel flow with $w_c = 0$, (*b*) counter flow with $w_c = 0$, and (*c*) parallel and counter flows with $w_c > 0$.

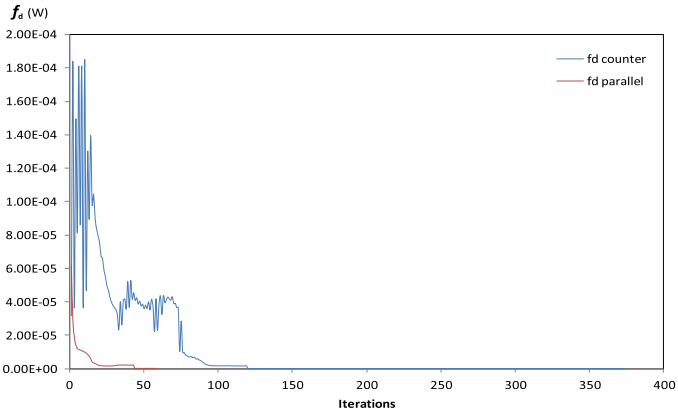


Figure 12. Minimization of fluid power dissipation with fluid phases separation, in parallel flow (red curve) and counter flow (blue curve).

pressure drop, becomes better than transporting the fluid in parallel straight pipe, regarding pressure drop minimization. For counter flow, in absence of continuity function (Figure 11b), the algorithm creates two fluid pipes that behave like a shortcut between inlets and outlets of different fluids on same side ($f_d = 1.4 \times 10^{-8}$ W). However for $w_c > 0$ continuity equation ensures that each fluid enters and leaves the domain from its preselected boundary. Hence two parallel pipes of fluid 1 and fluid 2 are created in parallel and counter flows, as seen in Figure 11c ($f_d = 9.4 \times 10^{-8}$ W). The continuity function penalized the objective function, which lead to optimal solutions that minimize pressure drop, at the condition that each fluid has its own continuous channel relating the predefined inlet and outlet boundaries sections.

In Figure 12, fluid power dissipation function f_d throughout the optimization process is sketched for the cases of parallel and counter flows including channels separation throughout the



(*a*) (*b*) (*c*)

Figure 13. Initial configuration (*a*) and topology optimization result without flow continuity for $w = 0.8$ (*b*) and $w = 0.3$ (*c*).

use of continuity objective function. The red curve (parallel flow) shows a smooth and fast convergence to the optimal solution. However, the blue curve (counter flow) shows an extremely high number of oscillations between iterations 1 and 80. The conflict between objective functions leads to these oscillations: fluid power dissipation function forces the fluid to take the shortcut path between inlet and outlet boundaries at the same edge, whereas continuity objective function builds an horizontal solid layer to forbid the fluid from taking the short path between boundaries that belong to different fluids subdomains. However after iteration 80, two horizontal pipes are created and the rest of iterations adjust the solid-fluid interface in each pipe to finally converge to two parallel and straight pipes.

4.2.2. Heat and mass transfer (w < 1)

The initial configuration for the topology optimization problem studied in this paragraph is depicted in Figure 13*a*. The dimensions of the domain are in mm, and the design grid size is 72×48. u and T stand respectively for fluid velocity and fluid terminal temperature. i, o, $f1$, and $f2$ stand respectively for inlet, outlet, fluid 1 and fluid 2. The velocity parabolic profile is similar on inlet and outlet boundaries of both fluids and the Reynolds number is $Re = 13$. Fluid 1 is considered the cold fluid, and fluid 2 as the hot fluid. The inlet temperatures of both fluids are $T_{i,f1} = 0$ °C and $T_{i,f2} = 150$ °C. Outlet fluid boundaries are characterized by a null temperature gradient. Fluid 1 and fluid 2 properties are the following: $\rho_{f1} = \rho_{f2} = 1000$ kg/m^3, $\mu_{f1} = \mu_{f2} = 0.001$ Pa s, $C_{p(f1)} = C_{p(f2)} = 4200$J/Kkg and $k_{f1} = k_{f2} = 1$W/kgm. Maximum fractions of both fluids are $\varphi_{f1} = \varphi_{f2} = 0.21$. Solid thermal conductivity is $k_s = 10$W/kgm.

Similarly to the previous paragraph, the problem is first solved without fluid separation ($w_c = 0$), whose corresponding topology optimization results are presented in Figure 13*b* for $w = 0.8$ and in Figure 13c for $w = 0.3$. The results showed two important features. First the creation of two parallel pipes between fluid 1 inlet and fluid 2 outlet sections and between fluid 2 inlet and fluid 1 outlet sections. Second, when w decreases, the pipes are driven toward south and north edges in order to increase solid thickness between the pipes, which minimizes heat transfer rate. In fact, it could be seen from Eq. (13), that maximization of f_e corresponds to the maximization of outlet temperature of fluid 1 and minimization of the outlet temperature of fluid 2. Normally $T_{o,f1}$ should correspond to the outlet temperature of the cold fluid ($T_{o,c}$) and $T_{o,f2}$ should correspond to the outlet temperature of the hot fluid ($T_{o,h}$). However the distribution of inlet and outlet boundaries of both fluids and the absence of continuity objective function that ensures that each fluid leaves the domain from its corresponding outlet boundary, allows the creation of structure similar to Figure 13b. Thus, there is a great advantage for the maximization of f_e, that hot fluid leaves the domain from outlet boundary of fluid 1, and the cold fluid leaves the domain from the outlet boundary of fluid 2. Furthermore, in this case increasing $T_{o,f1}$ and decreasing $T_{o,f2}$ requires minimization of heat transfer rate, hence in case of $w = 0.3$ the two pipes are driven away from each other, as seen in Figure 13b. However, in this case f_e does not represent the heat transfer rate between the fluids, and the absence of continuity function lead to an optimization problem that does not represent a real physical problem.

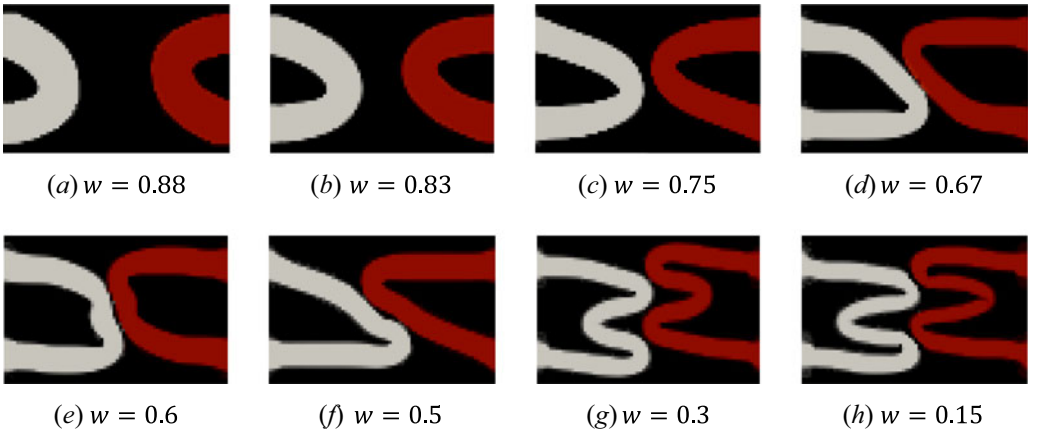


Figure 14. Topology optimization results with flow continuity.

Table 1. Thermal and hydraulic performance of topology optimization results.

w	f_d ($\times 10^{-7}$ W)	f_e($\times 10^{-4}$ W)	$T_{o,f1}$ (°C)	$T_{o,f2}$ (°C)	ΔT (°C)
0.88	0.674	2.88	18.37	131.64	113.27
0.83	1.134	3.43	21.07	128.94	107.87
0.75	2.396	4.53	27.09	122.94	95.85
0.67	5.181	6.01	30.86	120.37	89.51
0.6	5.383	6.11	38.13	111.42	73.29
0.5	6.241	6.31	41.19	108.01	66.82
0.3	17.24	7.08	43.79	105.79	62
0.15	32.94	9.83	49.57	102.23	52.66

Topology optimization results for a wide range of values of w with consideration of separation of fluids subdomains ($w_c > 0$) are presented in Figure 14. It can be noticed that a single pipe for each fluid is generated, connecting its inlet and outlet boundaries located on the same edge of the domain. Using continuity objective function ensured the convergence to a realistic physical solution, in which each fluid takes its own path independently from the second, hence f_e in this case represents the heat transfer rate between the streams. Thermo-hydraulic parameters of the optimal structures presented in Figure 14, are summarized in Table 1. An optimal heat exchanger can be summarized as follow: the geometrical configuration that provide the minimal pressure drop for a given heat transfer rate, or vice versa. In this problem, decreasing the weight w, means heat transfer term should be increased, which could be accomplished only at the detriment of pressure drop for optimal geometries, to get the minimum of the mutliobjective function F. When w decreases, the thickness of the solid domain separating the fluids is reduced by driving the pipes closer to each other. At the closet distance between the pipes, a single solid cell separates fluid 1 and fluid 2 cells. For $w = 0.3$ the residence time of fluid in the domain is increased, by increasing the total distance of the pipe in which the fluid is transported, which also increases heat transfer surface. The optimal structure of $w = 0.15$, is similar to the structure of $w = 0.3$ with lower diameter and higher total length, in order to increase furthermore heat transfer rate.

The variations of f_d and f_e throughout the optimization process are sketched in Figure 15 for $w = 0.5$, which optimal result is presented in Figure 14*f*. The initial guess solution is the optimal result that corresponds to the minimization of fluid power dissipation alone. The graph shows that f_e increases progressively until reaching a tradeoff solution between f_d and f_e for $w = 0.5$, which corresponds to the minimum of F, the penalty continuity function f_c being null. It can be noticed that for every infinitesimal increasing in f_e, corresponds an infinitesimal increasing in f_d, which is a normal and realistic physical behavior in optimization of heat exchangers, as discussed

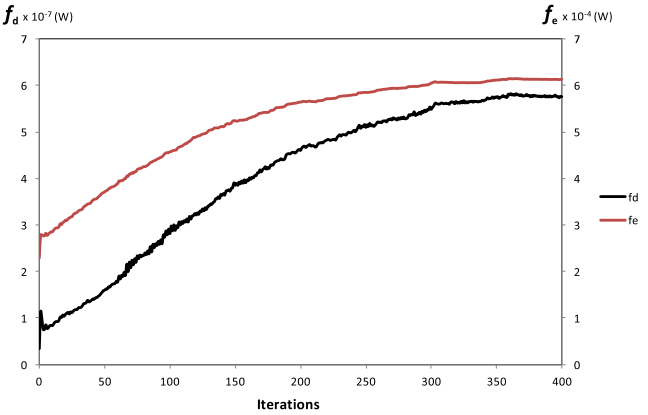


Figure 15. Variation of f_d and f_e throughout the optimization process for $w = 0.5$ with flow continuity.

previously. It should be noted that f_d and f_e have increased respectively by 1791% and 175% in relation to the optimal configuration in case of minimization of pressure drop alone ($w = 1$). This shows that pressure drop and heat transfer don't vary proportionally; in fact a small decreasing in outlet temperature difference between the fluids requires a much higher increasing in pumping power.

5. Conclusion

Topology optimization techniques are not based on predefined geometry, hence they allow reaching a complex and efficient configurations based only on the objectives and constraints. In the current state of scientific work, the application of topology optimization is limited to heat exchangers in which a single fluid exchanges heat with the solid structure. The present work aims to extend the topology optimization methods in fluid mechanics to cases including two fluids, which is the case for fluid-fluid heat exchangers. For this purpose, a new method in topology optimization of heat and mass transfer is introduced in this work, which can account for the optimal distribution of three different phases: two fluids and one solid. Multi-phase topology optimization was well studied and applied in mechanical structure problems, in which the goal was to account for the distribution of multiple materials in addition to the void phase. Bi-fluid topology optimization can be characterized by two main features that distinguish it from mono-fluid topology optimization:

- First, the bi-eta interpolation function, which was developed on the basis of the works on density approach applied on mono fluid flow problems, and the works on multi-phase topology optimization of mechanical structure problems.
- Second, the flow continuity function that acts like a penalty function that ensure flow continuity of each of fluid 1 and fluid 2 in the domain.

Both features were investigated by a set of case studies, some of them compared with results from the scientific literature of mono fluid topology optimization, bi-fluid topology optimization being not considered before, to the author knowledge. The results showed the capacity of the method to respond the bi-objective and bi-fluid nature of the problem. However, further investigations should look in applying the method on three dimensional domains, which is essential for the design and optimization of a new generation of heat exchangers, and also in considering turbulent flows and manufacturing constraints to avoid having innovative and efficient but non manufacturable configurations.

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